

create table items(itemid int primary key, Itemname varchar(50), category varchar(50), Price varchar(50), Instock int check (Instock >= 0));

desc items;

```
table ITEMS created.
desc items
Name      Null      Type
-----
ITEMID    NOT NULL  NUMBER(38)
ITEMNAME                      VARCHAR2(50)
CATEGORY                      VARCHAR2(50)
PRICE                        VARCHAR2(50)
INSTOCK                      NUMBER(38)
```

create table customers(custid int primary key, Custname varchar(30), Address varchar(50), state varchar(50));

desc customer;

```
table CUSTOMERS created.
.
Name      Null      Type
-----
CUSTID    NOT NULL  NUMBER(38)
CUSTNAME                      VARCHAR2(30)
ADDRESS                      VARCHAR2(50)
STATE                        VARCHAR2(50)
```

create table order_24(orderid int primary key, itemid int, foreign key(itemid) references items, custid int, foreign key(custid) references customers, quantity int, orderdate date);

desc order_24;

```
table ORDER_24 created.
desc order_24
Name      Null      Type
-----
ORDERID    NOT NULL  NUMBER(38)
ITEMID                      NUMBER(38)
CUSTID                      NUMBER(38)
QUANTITY                      NUMBER(38)
ORDERDATE                      DATE
```

create table delivery(deliverid int primary key, custid int, foreign key(custid) references customers, orderid int, foreign key(orderid) references order_24);

desc delivery;

```
table DELIVERY created.
```

```
desc delivery
```

```
Name          Null          Type
```

```
-----
```

```
DELIVERID NOT NULL NUMBER(38)
```

```
CUSTID          NUMBER(38)
```

```
ORDERID          NUMBER(38)
```

```
INSERT INTO Items VALUES (01, 'Samsung Galaxy S24', 'Electronics', 79999, 50);
```

```
INSERT INTO Items VALUES (02, 'Apple iPhone 15', 'Electronics', 89999, 30);
```

```
INSERT INTO Items VALUES (03, 'Dell ', 'Computers', 65000, 20);
```

```
INSERT INTO Items VALUES (04, ' Headphones', 'Accessories', 29999, 40);
```

```
INSERT INTO Items VALUES (05, 'Water Bottle', 'Home ', 32000, 15);
```

```
select * from items;
```

	ITEMID	ITEMNAME	CATEGORY	PRICE	INSTOCK
1	1	Samsung Galaxy S24	Electronics	79999	50
2	2	Apple iPhone 15	Electronics	89999	30
3	3	Dell	Computers	65000	20
4	4	Headphones	Accessories	29999	40
5	5	Water Bottle	Home	32000	15

```
INSERT INTO Customers values(101,'Ann','Kottayam','Kerala');
```

```
INSERT INTO Customers values(102,'Mariya','Pala','Kerala');
```

```
INSERT INTO Customers values(103,'Appu','Muvathupuzha','Kerala');
```

```
INSERT INTO Customers values(104,'John','Ettumanoor','Kerala');
```

```
INSERT INTO Customers values(105,'Mickey','Kothamangalam','Kerala');
```

```
select * from customers;
```

	CUSTID	CUSTNAME	ADDRESS	STATE
1	101	Ann	Kottayam	Kerala
2	102	Mariya	Pala	Kerala
3	103	Appu	Muvathupuzha	Kerala
4	104	John	Ettumanoor	Kerala
5	105	Mickey	Kothamangalam	Kerala

```

INSERT INTO Order_24 VALUES (1001, 01, 101, 1, TO_DATE ('2026-01-11','YYYY-MM-DD'));
INSERT INTO Order_24 VALUES (1002, 03, 102, 1, TO_DATE ('2026-01-10','YYYY-MM-DD'));
INSERT INTO Order_24 VALUES (1003, 04, 103, 2, TO_DATE ('2026-01-15','YYYY-MM-DD'));
INSERT INTO Order_24 VALUES (1004, 05, 105, 1, TO_DATE ('2026-01-10','YYYY-MM-DD'));
INSERT INTO Order_24 VALUES (1005, 02, 104, 1,TO_DATE ('2026-01-10','YYYY-MM-DD'));
INSERT INTO Order_24 VALUES (1006, 05, 105, 1, TO_DATE ('2026-01-10','YYYY-MM-DD'));
select * from order_24;

```

R2	ORDERID	R2	ITEMID	R2	CUSTID	R2	QUANTITY	R2	ORDERDATE
1	1001		1		101		1	11-01-26	
2	1002		3		102		1	10-01-26	
3	1003		4		103		2	15-01-26	
4	1004		5		105		1	10-01-26	
5	1005		2		104		1	10-01-26	
6	1006		5		105		1	10-01-26	

```

INSERT INTO Delivery VALUES (5001, 105, 1006);
INSERT INTO Delivery VALUES (5002, 102, 1002);
INSERT INTO Delivery VALUES (5003, 103, 1003);
INSERT INTO Delivery VALUES (5004, 104, 1005);
select * from delivery;

```

R2	DELIVERID	R2	CUSTID	R2	ORDERID
1	5001		105		1006
2	5002		102		1002
3	5003		103		1003
4	5004		104		1005

1) SELECT * FROM Customers JOIN Order_24 ON Customers.custid = Order_24.custid;

R2	CUSTID	R2	CUSTNAME	R2	ADDRESS	R2	STATE	R2	ORDERID	R2	ITEMID	R2	CUSTID_1	R2	QUANTITY	R2	ORDERDATE
1	101	Ann	Kottayam	Kerala		1001		1	101		1	11-01-26					
2	102	Mariya	Pala	Kerala		1002		3	102		1	10-01-26					
3	103	Appu	Muvathupuzha	Kerala		1003		4	103		2	15-01-26					
4	105	Mickey	Kothamangalam	Kerala		1004		5	105		1	10-01-26					
5	104	John	Ettumanoor	Kerala		1005		2	104		1	10-01-26					
6	105	Mickey	Kothamangalam	Kerala		1006		5	105		1	10-01-26					

- 2) SELECT DISTINCT c.* FROM Customers c JOIN Delivery d ON c.Custid = d.Custid;

	CUSTID	CUSTNAME	ADDRESS	STATE
1	102	Mariya	Pala	Kerala
2	105	Mickey	Kothamangalam	Kerala
3	103	Appu	Muvathupuzha	Kerala
4	104	John	Ettumanoor	Kerala

- 3) SELECT c.Custname , o.orderdate FROM Order_24 o JOIN Customers c ON o.Custid = c.custid WHERE c.Custname LIKE 'J%';

	CUSTNAME	ORDERDATE
1	John	10-01-26

- 4) SELECT DISTINCT i.Itemname, Price FROM Items i JOIN Order_24 o ON i.itemid = o.itemid JOIN Customers c ON o.Custid = c.Custid WHERE c.Custname = 'Mickey';

	ITEMNAME	PRICE
1	Water Bottle	32000

- 5) SELECT DISTINCT c.*FROM Customers c JOIN Order_24 o ON c.CustID = o.CustID LEFT JOIN Delivery d ON o.Orderid = d.Orderid WHERE o.orderdate > '31-01-2013' AND d.deliverid IS NULL;

	CUSTID	CUSTNAME	ADDRESS	STATE
1	101	Ann	Kottayam	Kerala
2	105	Mickey	Kothamangalam	Kerala

- 6) SELECT itemid FROM Order_24 UNION SELECT o.itemid FROM Order_24 o LEFT JOIN Delivery d ON o.Orderid = d.Orderid WHERE d.Deliverid IS NULL;

	ITEMID
1	1
2	2
3	3
4	4
5	5

- 7) SELECT Custname FROM Customers WHERE CustID IN (SELECT CustID FROM Order_24)
INTERSECT SELECT Custname FROM Customers WHERE CustID IN (SELECT o.CustID FROM
Order_24 o JOIN Delivery d ON o.Orderid = d.Orderid);

	CUSTNAME
1	Appu
2	John
3	Mariya
4	Mickey

- 8) SELECT Custname FROM Customers WHERE CustID IN (SELECT CustID FROM Order_24)
MINUS SELECT Custname FROM Customers WHERE CustID IN (SELECT o.CustID FROM
Order_24 o JOIN Delivery d ON o.Orderid = d.Orderid);

	CUSTNAME
1	Ann

- 9) SELECT Custname FROM Customers WHERE CustID = (SELECT CustID FROM Order_24 GROUP
BY CustID ORDER BY COUNT(Orderid) DESC FETCH FIRST 1 ROW ONLY);

	CUSTNAME
1	Mickey

- 10) SELECT DISTINCT c.* FROM Customers c JOIN Order_24 o ON c.CustID = o.CustID JOIN Items i
ON o.itemid = i.itemid WHERE i.Price > 5000;

	CUSTID	CUSTNAME	ADDRESS	STATE
1	102	Mariya	Pala	Kerala
2	101	Ann	Kottayam	Kerala
3	105	Mickey	Kothamangalam	Kerala
4	103	Appu	Muvathupuzha	Kerala
5	104	John	Ettumanoor	Kerala

- 11) SELECT Custname, address FROM Customers WHERE CustID NOT IN (SELECT o.CustID FROM
Order_24 o JOIN Items i ON o.itemid = i.itemid WHERE i.Itemname = 'Samsung Galaxy S24');

	CUSTNAME	ADDRESS
1	Mariya	Pala
2	Appu	Muvathupuzha
3	John	Ettumanoor
4	Mickey	Kothamangalam

12)

Left Outer Join: `SELECT c.*, o.* FROM Customers c LEFT OUTER JOIN Order_24 o ON c.CustID = o.CustID;`

	CUSTID	CUSTNAME	ADDRESS	STATE	ORDERID	ITEMID	CUSTID_1	QUANTITY	ORDERDATE
1	101	Ann	Kottayam	Kerala	1001	1	101	1	11-01-26
2	102	Mariya	Pala	Kerala	1002	3	102	1	10-01-26
3	103	Appu	Muvathupuzha	Kerala	1003	4	103	2	15-01-26
4	105	Mickey	Kothamangalam	Kerala	1004	5	105	1	10-01-26
5	104	John	Ettumanoor	Kerala	1005	2	104	1	10-01-26
6	105	Mickey	Kothamangalam	Kerala	1006	5	105	1	10-01-26

Right Outer Join:

`SELECT c.*, o.* FROM Customers c RIGHT OUTER JOIN Order_24 o ON c.CustID = o.CustID;`

	CUSTID	CUSTNAME	ADDRESS	STATE	ORDERID	ITEMID	CUSTID_1	QUANTITY	ORDERDATE
1	101	Ann	Kottayam	Kerala	1001	1	101	1	11-01-26
2	102	Mariya	Pala	Kerala	1002	3	102	1	10-01-26
3	103	Appu	Muvathupuzha	Kerala	1003	4	103	2	15-01-26
4	104	John	Ettumanoor	Kerala	1005	2	104	1	10-01-26
5	105	Mickey	Kothamangalam	Kerala	1006	5	105	1	10-01-26
6	105	Mickey	Kothamangalam	Kerala	1004	5	105	1	10-01-26

13) `SELECT state, COUNT(*) AS total_customers FROM Customers GROUP BY state;`

STATE	TOTAL_CUSTOMERS
1 Kerala	5

14) SELECT * FROM Items WHERE Price > (SELECT AVG(Price)
FROM Items) ORDER BY category;

	ITEMID	ITEMNAME	CATEGORY	PRICE	INSTOCK
1	3	Dell	Computers	65000	20
2	1	Samsung Galaxy S24	Electronics	79999	50
3	2	Apple iPhone 15	Electronics	89999	30